Federated Edge-assisted Mobile Clouds for Service Provisioning in Heterogeneous IoT Environments

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***Abstract*—In this paper, we present a novel hybrid approach, where Fog computing features are used to support a dynamic cloud cooperation of mobile IoT devices. In particular, we introduce the so-called Mobile-IoT-Federation-as-a-Service (MIFaaS) paradigm, according to which edge nodes operate as orchestrators to manage federations among public/private IoT clouds that enable integrated IoT applications. To this aim, we propose an algorithm to foster the federation of local IoT clouds based on an utility function accounting for the number of executed tasks to be maximized. The presented performance evaluation validates the enhancements achievable by the proposed solutions in terms of number of task requests being successfully executed.**

***Index Terms*—IoT, Fog Computing, Mobile Cloud, Cloud federation**

I. Introduction

In the last decade a plethora of Cloud platforms have been developed for advanced Internet of Things (IoT) services. IoT devices are connected to cloud servers where their virtualized image enables remote service provisioning regardless of the computation capabilities of the physical objects. Such a cloud centric vision for worldwide IoT implementations received growing endorsements by research and industry communities promoting the interaction among private and public IoT clouds. In this context, Fog computing [1] allows to meet the users' requirements of faster processing time and low latency by promoting the idea of using a distributed virtualization platform. This is achieved at the cost of deploying a large number of servers, located at the edge of the network between end-user devices and the cloud data centers.

On the other side, opportunistic cooperation among IoT devices is an investigation field gaining even more popularity. The underlying idea for this approach, traditionally falling within the Mobile Cloud Computing field [2], is to exploit local connectivity to create a dynamic cloud. In particular, devices can cooperatively reach their goal of reduced service execution time and lower energy consumption. Mobile Cloud Computing (MCC) is a well-investigated field. Notwithstanding, bridging this concept to the IoT paradigm [3] introduces several challenges related to: (i) *virtualization* of heterogeneous resources (in addition to the typical cloud services, i.e., computation, storage and networking, also sensing, actuation, and other location-based services are required); (ii) *variety of IoT hardware platforms* ranging from autonomous and high performing vehicles to resource-constrained sensors; (iii) *heterogeneity* of the technologies and communication solutions; (iv) *security* issues, related to the exchange of privacy information among devices. Moreover, completely distributed approaches for task allocation typically present not affordable requirements for IoT devices leading to *scalability and reliability* issues in a highly dynamic environment.

In this paper, we present a novel hybrid approach, where Fog computing features are used to support a *dynamic cloud federation* among mobile IoT devices. Our solution allows for better scalability compared to a micro data center deployed in the network access point, a so-called cloudlet, since resources can be provisioned on-the-fly by the involved IoT devices and the deployment cost of micro data center at the edge of the network is reduced. The proposed architecture promotes the federation of multiple private/public IoT clouds, the so-called IoT Cloud Providers (ICPs), which can share the resources of their own devices to effectively implement high level services and, thus, increase the number of executed tasks. Different from classic collaborative strategies in MCC literature, where the single device can establish opportunistic cooperation only exploiting its own resources, in our approach the federation process considers the whole pool of devices managed by each involved ICP to extend the opportunities of reciprocal support in a heterogeneous IoT scenarios. In particular, we introduce the so-called Mobile-IoT-Federation-as-a-Service (*MIFaaS*) paradigm, where the edge node operates as an orchestrator to manage federations among ICPs to run integrated IoT applications.

To point out the potential benefits of the proposed approach for private/public actors, we consider a simple use case related to intelligent transportation system (ITS) services. John has recently bought his smart car, which can be connected to his local personal cloud composed by his other personal devices, such as his smartphone, tablet and smart watch. Through his cloud, John can monitor his travels and evaluate the fuel consumption as well as exploit the multimedia service offered by his virtualized movie catalog to entertain his children during the travel. After a few weeks, he finds himself stuck in a traffic jam and his cloud receives a request to join a federation of private and public local clouds. The municipality ICP, which has triggered the MIFaaS federation formation, can only produce information about the traffic status through cameras alongside the road, but lacks of computation and storage capabilities. Therefore, it queries the edge node to look for other ICPs able to help in processing the necessary operations to re-schedule the traffic lights in real-time and

to resolve the traffic jam. Unfortunately, John's car, which is clearly interested in the ITS service, is already executing other high computation multimedia services and is unable to simultaneously handle multiple services at a time. With the proposed paradigm, the edge node will decide which other available devices of John's cloud can provide support to resolve the traffic jam tasks instead.

By providing potential unlimited computation, storage, sensing and actuation capabilities, the number and type of possible applications enabled by our paradigm are boundless and can involve each aspect of human life. However, to reach this goal we should cope with new challenges related to the federation of mobile IoT clouds. The main contributions in this paper can be summarized as it follows:

- introduction of a novel Mobile-IoT-Federation-as-a-Service (MIFaaS) paradigm to promote the dynamic cooperation of ICPs in heterogeneous IoT environments;
- proposal of an algorithm to foster the federation of ICPs and definition of a utility function to be maximized accounting for the number of tasks executed;
- performance evaluation showing the enhancements obtained in terms of satisfied tasks with the proposed approach.

The paper is organized as follows. Section II provides an overview on the related works. Section III illustrates the reference system and the basic components of the proposed MIFaaS paradigm. In Section IV the problem of the edge-assisted IoT cloud provider federation is formulated defining a federation formation algorithm. Finally, the performance evaluation is reported in Section V, whereas conclusions and future research hints are given in VI.

II. Related Work

The literature related to MCC [2] has extensively investigated approaches of opportunistic interaction between mobile devices to provide enough resources for the desired service. Examples of cooperative solutions to create local mobile cloud are available in [4], [5] and [6]. These solutions foresee distributed cooperation mechanism to create collaborative computing platforms to share the resources of the involved devices. The main drawback of this distributed approach is the signaling overhead to coordinate cooperation and provide the involved devices with an up-to-date synchronized view of the current network state, available resources, and received service requests. On the contrary, an infrastructure-based approach enables task offloading from mobile devices to remote high-capability data centers, as proposed in [7], [8] and [9]. Typical limitations, in terms of latency and bandwidth requirements for communication with a remote server, are overcame by deploying micro data centers in the network access points, known as Cloudlets, as proposed in [10]. Bringing the cloudlet concept into the IoT scenario results in the definition of the so-called Fog Computing [11], where specific IoT-related challenges are accounted for, such as the huge number of heterogeneous IoT devices. However, according to this approach, telco providers are facing the issue of a good balancing between resource dimensioning to support the services and the financial costs to deploy a large number of processing nodes located very close to clients.

Our proposal of a hybrid paradigm of *Federated Edge-assisted Mobile clouds (FEM-clouds)* can support telco providers in achieving the advantages of both Fog computing and distributed cooperation among mobile devices, overcoming their respective drawbacks. In particular, the proposal exploits the opportunistic resources available from multiple ICPs in the local edge node area and at the same time enables scalability features. In fact, edge nodes are designed for control and orchestration functions, thus reducing the needed processing capabilities, whereas task execution is completely delegated to IoT devices. The idea of a resource coordinator elected among the mobile nodes is also proposed in [12] to manage the matching between application requests and device resources. In [13] instead, the authors propose the refactoring of the cloudlet in a controller so that nearby devices with sufficient stability can be configured into a coordinated cloud computing service. However, both papers do not actually tackle the heterogeneity of IoT devices in terms of power, autonomy, and capabilities. Moreover, they consider a single unique mobile local cloud; thus their solution might suffer from scalability and flexibility issues. The key expected advantage of our approach is to extend the model of cooperation typically presented for MCC, where the single device can receive support from near opportunistic devices by exposing its own resources only. In our system, the focus is moved to the ICPs and to the whole sets of their devices capabilities and application requests to create a more flexible and integrated IoT service paradigm. A further aspect addressed in this paper is task allocation for IoT. Although task mapping has been extensively analyzed in the literature [14], IoT introduces several constraints and challenges which require a new appropriate problem formulation. To this aim, different heuristics have recently been proposed, e.g., [15] and [16]. On the other hand, an emerging trend in the Cloud literature deals with cooperative approaches for distributing workload among multiple cloud providers [17] [18]. In this paper, we extend these approaches for dynamic IoT scenarios and design appropriate federation schemes for ICPs, which share the resources of their own devices on-the-fly.

III. Reference System and Architecture

In this paper, we assume that objects are organized into ICPs, either public (e.g., groups of smart devices belonging to a municipality) or private (e.g., objects owned by a user). We introduce the concept of *Mobile-IoT-Federation-as-a-Service (MIFaaS)* according to which the edge node operates as an orchestrator to manage federations of public/private ICPs. The role of the edge node may be played by either the telco or any other entity who may have a business interest in providing additional and enhanced services to the service subscribers, such as a coffee shop owner, a university, or a smart city in general. Thus, the edge node identifies the atomic tasks required to run integrated IoT applications, through the use of

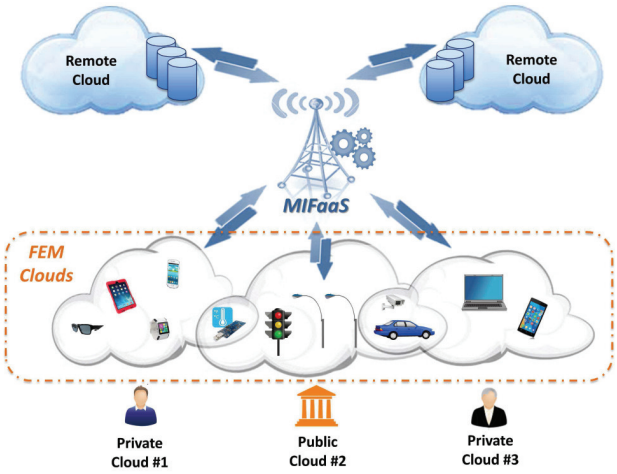

Fig. 1. FEM Clouds scenario

a Decomposer module, and then decides the allocation of the needed resources among the objects of the local ICPs. The goal of an ICPs federation is to collect enough capabilities from the objects to run the application, according to the desired user requirements and their corresponding tasks (e.g., computing, storage and data sensing). If the resources collected from the federation are not enough to complete the designed operations, then the task execution could be supported by remote cloud servers. Under this view, our proposal can be considered as complementary to traditional task offloading to a remote server and aims at providing better services, by exploiting opportunistic cooperation among heterogeneous local clouds. An example of the defined scenario is depicted in Figure 1, where different private/public ICPs are illustrated. For instance, users can exploit the potentials offered by their own clouds, made up of the resources provided by their local devices. Furthermore, municipalities in a Smart City deploy their clouds of objects, such as roadside units, security cameras, etc., to offer public services to citizens. In this heterogeneous context of mobile and static IoT clouds, the edge node acts as a broker supporting the creation and maintenance of opportunistic federations for enabling enhanced IoT services. In particular, we define a *federated mobile IoT cloud* as the federation of multiple autonomous ICPs finalized to cooperatively offer IoT services by combining the capabilities of their heterogeneous devices.

When a user requires a composite IoT application, if his/her cloud of devices does not have enough capabilities to handle the application, it sends a resource request to the edge node. After having collected the requests over a small period of time, the edge node triggers the federation formation process by virtualizing the computational, communication, storage, and information resources. These resources are managed by the edge node, so that every task can be assigned to the fittest object based on the capabilities available by each object. We highlight that, once the federation has been decided, then the edge node notifies the requesting ICP about the list of supporting ICPs, by specifying the number and type of the tasks to be offloaded. This approach allows to deploy tasks across different federated ICPs devices, thus applications are completely agnostic of the underlying physical devices and could be migrated in part or as a whole between cooperating ICPs. To enable this flexibility, appropriate virtualization techniques are required to provide the required hardware abstraction and to develop an interoperable service platform for heterogeneous IoT devices. Each ICP can offer its resources to one or more different ICPs. This is the example of the information provided by a sensor, which can be useful to several FEM clouds. Therefore, it is of utmost importance to establish a mechanism to wisely allocate the tasks. To this aim, the edge node provides support in the creation and management of the federation among clouds, by mediating the requests received by the ICPs and facilitating the synchronization status of the resources offered by each device. With respect to the Fog computing scenario, this solution helps in deploying edge nodes with lower costs for the provider since they mainly have management responsibilities.

IV. Service Model for the Edge-assisted IoT Cloud Provider Federation

With reference to the solutions presented in the previous section, we consider a scenario with a single edge node and several mobile ICPs. The main objective of the edge node is to create and orchestrate one or more federations of ICPs to increase the amount of services offered. We assume that the edge node collects over a certain time frame the service requests coming from the ICPs in the area under its control and then, based on the information about the available resources, assists the ICPs in creating well performing ICP federations.

Let us define a set of N ICPs $\mathcal{N} = \{I_1, I_2, \ldots, I_N\}$, where each ICP i (either private or public) is endowed with a set of $\mathcal{D}_i$ devices. A generic device d is capable of performing tasks according to the set of associated resources $\mathcal{R}_{i,d} = \{R^m : m \in \mathcal{M}\}$, where m indicates the type of resource selected from the set of possible resource types $\mathcal{M}$ (computation, different types of sensing, storage, etc.). A further important feature characterizing an ICP is its mobility. In particular, an ICP can be either *static*, *mobile with small mobility* (i.e., pedestrian), or *mobile with high mobility* (i.e., vehicular). The mobility parameter is of high importance to evaluate the probability that a task is performed and its result is effectively received by the requesting ICP. In fact, we assume that once the tasks are allocated to an ICP, the results are sent locally to the requesting ICP without passing through the edge node. Therefore, both requesting and executing ICP need to remain under mutual coverage for the time requested to execute the task and transfer the results. To model this aspect, in our problem formulation we introduce a mobility factor $p_i \in [0, 1]$, where zero stands for a static ICP, whereas one stands for an ICP moving away with probability one from the communication range (causing a loss in the results for the allocated tasks).

A single ICP i has a workload consisting of a set of application requests $\mathcal{A}_i$. Each application request a_i is mapped by the Decomposer in the edge node to a set of elementary

task requests[1], with $t^m_{i,a}$ being a single task of type m and $\mathcal{T}^a_i$ the total set of tasks related to application request a_i from ICP i. In the service model, for simplicity, we assume that the edge node has a predefined time frame to collect all the service requests and that each requested task can be completed in this time frame. Moreover, in modeling the problem we introduce a distinction in the tasks: *exclusive use task*, that is when a task result can be used by a single ICP only; and *shared use task* when a task result can be useful for multiple requests coming from different ICPs. Computation is an example of the first category of tasks, since the relevant physical resources can be usually allocated to a single specific request, whereas sensing is a typical representative of the second category, as the same sensed information can be exploited by different applications. To this aim, we introduce a so-called *reuse factor* associated to a task t: $\mu_t \in \{0, 1\}$[2].

A. ICP federation algorithm

Once the edge node has collected all the metadata information from the N ICPs about the set of devices capabilities and the set of task requests, the next step for the definition of possible federations is to map the service requests to the available resources according to a *task allocation algorithm*. This will define the number of tasks requests actually being served for the involved ICPs. To this aim, we define a *utility* function that measures the ability of any combination of ICPs to maximize the number of solved tasks.

Algorithm 1: Mapping between task requests and devices

Data: $\mathcal{S}$, $\mathcal{D}_i$, $\mathcal{A}_i$, $\mathcal{R}_{i,d} \forall i \in \mathcal{S}$ and $\forall d \in \mathcal{D}_i$, $\mathcal{T}^a_i \forall i \in \mathcal{S}$ and $\forall a_i \in \mathcal{A}_i$
Result: Set of served applications $\mathcal{SA}$

$\mathcal{AQ} = \bigcup_{i \in \mathcal{S}} \mathcal{A}_i$
$\mathcal{TQ} = \bigcup_{i \in \mathcal{S}} \bigcup_{a_i \in \mathcal{A}_i} \mathcal{T}^a_i$ ▹ Set of task requests in federation $\mathcal{S}$
$\mathcal{SA} = \emptyset$ ▹ Set of finally served applications
$\mathcal{ST} = \emptyset$ ▹ Set of finally served tasks
for all $t \in \mathcal{TQ}$ **do**
  if $(\mu_t == 1)$ **then**
    if $(\exists t' \in \mathcal{ST}$ *that is equivalent)* **then**
      $\mathcal{ST} = \mathcal{ST} \cup t$
    else
      Select device d with resources for task t;
      Update resources for the device d;
      $\mathcal{ST} = \mathcal{ST} \cup t$
    end
  else
    Select device d with resources for task t;
    Update resources for the device d;
    $\mathcal{ST} = \mathcal{ST} \cup t$
  end
end
for all $a_i \in \mathcal{AQ}$ **do**
  if $(\mathcal{T}^a_i \subset \mathcal{ST})$ **then**
    $\mathcal{SA} = \mathcal{SA} \cup a_i$
  end
end

[1]Noteworthy, a task request type and an available resource in an ICP are identified by the same set $\mathcal{M}$.

[2]A possible extension could consider $\mu_t \in [0, 1]$, when a task is partially sharable

In Algorithm (1) the steps for the proposed task allocation solution implemented by the edge node are listed. In particular for every subset of ICPs $\mathcal{S} \subset \mathcal{N}$, the input parameters for the algorithm are, from the one side, the set of devices $\mathcal{D}_i$ associated to each ICP $i \in \mathcal{S}$ with the corresponding resources for each device $d \in \mathcal{D}_i$, i.e., $\mathcal{R}_{i,d}$, and, from the other side, the set of application requests $\mathcal{A}_i$ with the relevant tasks $t^m_{i,a} \in \mathcal{T}^a_i$ corresponding to each application request a^m_i from ICP $i \in \mathcal{S}$. The objective of the proposed task allocation algorithm is to maximize the number of served tasks. In particular, when a new task from the list $\mathcal{TQ}$ needs to be allocated, the edge node looks for a suitable device (i.e., it has the needed resources). If multiple devices are available, the edge node chooses the node which is currently serving the lowest number of tasks so that a fair load balancing is implemented. Moreover, if the reuse factor is equal to one, it also verifies if a similar task has already been served by the federation so that no additional resources are being allocated.

Once the task allocation has been performed for a given set of ICPs, we are able to evaluate the number of tasks being successfully executed by the considered federation. The selection of the best configuration of federated ICPs is a typical coalition formation problem. Being N the number of ICPs that can potentially federate, we will have $2^N - 1$ possible federations that can be formed. For each of these possible federations we associate a single value that gives the measure of how preferred this is w.r.t. the other federations. To this aim, let $U(\mathcal{S})$ be the utility associated to cooperative federation $\mathcal{S}$. Once the computation of the utility for any of the possible coalitions is performed, the edge node is able to select the best groups of cooperating ICPs. In particular, the edge node ranks all possible federations in a decreasing order of the associated $U(\mathcal{S})$ and selects the first value as this is guaranteeing the *maximum utility*. If this first elected coalition is not the *grand coalition* (i.e., all available ICPs are involved), then the remaining ICPs will be considered for a new step of the coalition selection algorithm until all the ICPs belong at least to a federation, where also *singleton* federations are considered (i.e., federations with one single non cooperating ICP).

The proposed utility function for a federation of $\mathcal{S}$ ICPs considers the total number of application requests being served. In particular, an application request a_i is considered as served if and only all the atomic tasks the Decomposer associates to the application request, i.e., $t^m_{i,a} \in \mathcal{T}^a_i$, are being executed. The success probability for the application request is given by the intersection of the success probabilities for the single tasks, which are independent events and thus yields the product of the success probabilities of the single tasks forming the application request: $P_s(a_i) = \prod_{t^m_{i,a} \in \mathcal{T}^a_i} P(t^m_{i,a})$. In our problem, we will consider for each ICP the sum of the number of tasks executed by itself and the number of tasks delegated to other ICPs in the federation. When a task is executed by the ICP itself the success probability is equal to one. Instead, for a task result to be successfully received by an ICP when this is

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Number of ICPs	[3-12]
Mobility factor per ICP	0.2
Number applications per ICP	5
Computation units per task request	[1-10] MFLOPS
Storage units per task request	[100-1000] MBytes
Number of sensing types	4
Sensing units per task request	[1-10]
Number of devices per ICP	5
Computation resource units per device	[0-10] MFLOPs
Storage resource units per device	[0-1000] MByte
Sensing resource units per device	[0-1]

executed by another ICP in the federation, the condition is that none of the two ICPs leaves the area under control of the edge node. The mobility factor can be treated as two stochastically independent events as the occurrence of one does not affect the probability of the other. Consequently, the success probability (i.e., the result is successfully completed and received by the requesting ICP) of a task requested by ICP i and allocated to ICP j equals to the product of the probabilities of the two ICPs staying in the area covered by the edge node. Summarizing we have that given a task $t^m_{i,a}$ the success probability of the task being executed by ICP j is defined as:

$$P(t^m_{i,a}) = \begin{cases} 1 & i = j \\ (1-p_i)(1-p_j) & i \neq j \end{cases}$$

The resulting utility associated to a federation $\mathcal{S}$ of S ICPs is the result of the following maximization problem:

$$U(\mathcal{S}) = \max \left\{ \frac{\sum_{i=1}^{S} \sum_{a_i \in \mathcal{A}_i} \left[\left(\sum_{t^m_{i,a} \in \mathcal{T}^a_i} \sum_{j=1}^{S} x_{t,a,i,j} \right) \cdot P_s(a_i) \right]}{\sum_{i=1}^{S} \sum_{a_i \in \mathcal{A}_i} |\mathcal{T}^a_i|} \right\} \quad (1)$$

where $P_s(a_i)$ is the success probability for the application request a_i (which is given by the product of the success probabilities of the single tasks as discussed earlier), $x_{t,a,i,j}$ is a variable assuming value 1 if task t being part of application request a_i, requested by ICP i, is implemented by ICP j (note that i may be equal to j if task t is implemented by the requesting ICP itself) and value 0 otherwise. Note that the constraints on the available resources for the devices, and the exploitation of the reuse possibilities for the requested task results are implicitly considered in the task allocation algorithm.

V. PERFORMANCE EVALUATION

In this section we evaluate the performance of the proposed MIFaaS paradigm in terms of number of executed tasks. We consider a generic IoT scenario where ICPs can request several application services, whose elementary tasks include computation, storage, and sensing activities. For the sake of simplicity, we assume that each application service foresees tasks of a predefined type. Requests for computation and storage tasks accounts for the typical available capacity in real smart mobile devices, as reported in [13]. Whereas, we consider four different possible types of sensing tasks (each related to a different sensed data). Sensing requests can also include multiple data sensing samples relevant to the same information, to guarantee the desired Quality of Information.

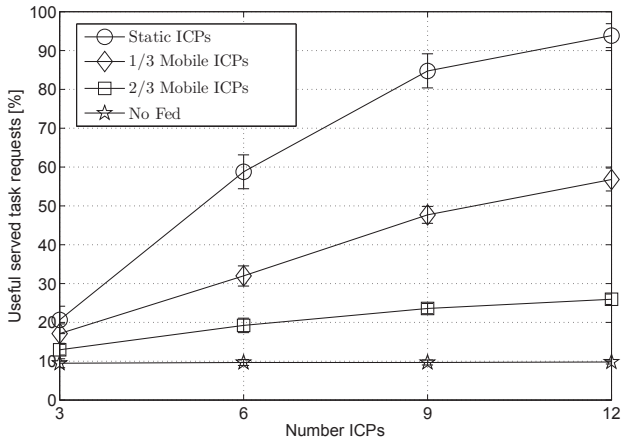

Fig. 2. Percentage of useful executed tasks.

Similarly, each ICP has a given number of heterogeneous devices, whose amounts of resources can range within intervals similar to those defined for the application tasks. Furthermore, each ICP may present different mobility profile, which is characterized by specifying a mobility factor p_i, as defined in Section IV. Table I summarizes our simulation parameters.

The analysis is conducted by using a simulation code developed in Matlab to study the impact of the number of ICPs and their mobility on the overall performance. In particular, we analyze the results of the proposed federation algorithm when considering four scenarios: (i) all the ICPs are static; (ii) one third of the ICPs are mobile, and their mobility parameter is equal to 0.2; (iii) two third of the ICPs are mobile with mobility parameter set to 0.2; (iv) ICPs are not willing to federate and work as stand-alone ICPs. To compute an accurate confidence interval, each scenario has been evaluated over 40 different runs. The results plotted in Fig. 2 refer to the percentage of useful executed tasks (please note that any task is considered as successful only if all the tasks of the same application request have been executed). As we can observe from the plot, the proposed solution based on ICP federations can achieve higher values of percentage of useful executed tasks. The performance level increases with the number of ICPs, since a greater amount of available resources allows to satisfy a higher number of application services; a value higher than 90% is achieved in the case of 12 static ICPs. As expected, the increase in the number of ICPs does not influence the percentage of executed task in a non-cooperative scenario, since each ICP can rely on its own devices only. As a last observation, in the scenarios with more mobile ICPs we

notice a reduction in the performances w.r.t. the static situation since mobile ICPs move away from the communication range with the other cooperating ICPs and, thus, are prevented from exchanging the results of the assigned tasks.

VI. Conclusions and Future works

In this paper we have defined a novel paradigm, the so-called Mobile-IoT-Federation-as-a-Service (MIFaaS), where the edge node operates as an orchestrator to manage federations among public/private IoT clouds to run composite IoT applications. Furthermore, we have proposed an algorithm to foster the federation of ICPs, while introducing an appropriate utility function to maximize the number of executed tasks. The presented performance evaluation has assessed that the proposed solution allows for significant increases in the number of served tasks. In future works, we planned to further improve our ICP federation algorithm to account for the device energy levels, to perform ICP mobility prediction to better allocate the requested tasks, and to put attention on issues of fairness among the cooperating ICPs.

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